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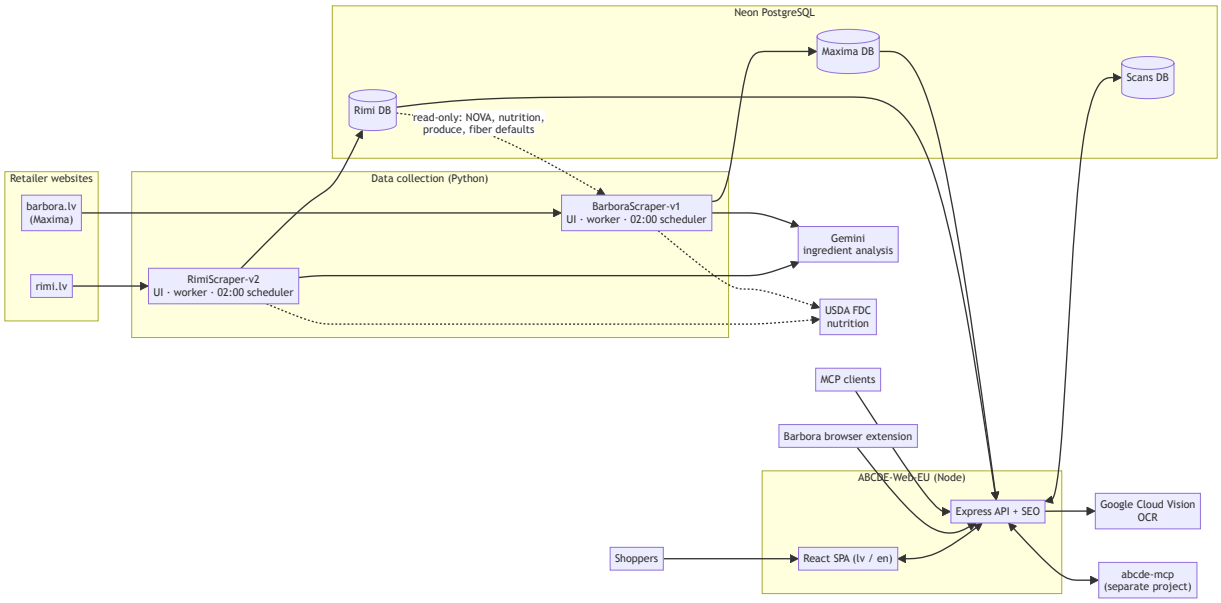
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ABCDE System Architecture

This document shows how the three repositories in this folder work together to run abcde.lv. Each repository has its own `ARCHITECTURE.md` with more detail:

Repository	Role	Stack	Detail
RimiScraper-v2	Collects and scores Rimi products	Python, Streamlit, PostgreSQL, Gemini	Job-queue worker, Nutri-Score/NOVA formulas
BarboraScraper-v1	Collects and scores Maxima products (Barbora.lv)	Python, Streamlit, PostgreSQL, Gemini	Fork of the Rimi scraper, linked to Rimi data
ABCDE-Web-EU	Public website and API	TypeScript, React, Express 5, Drizzle	Reads both scraper databases

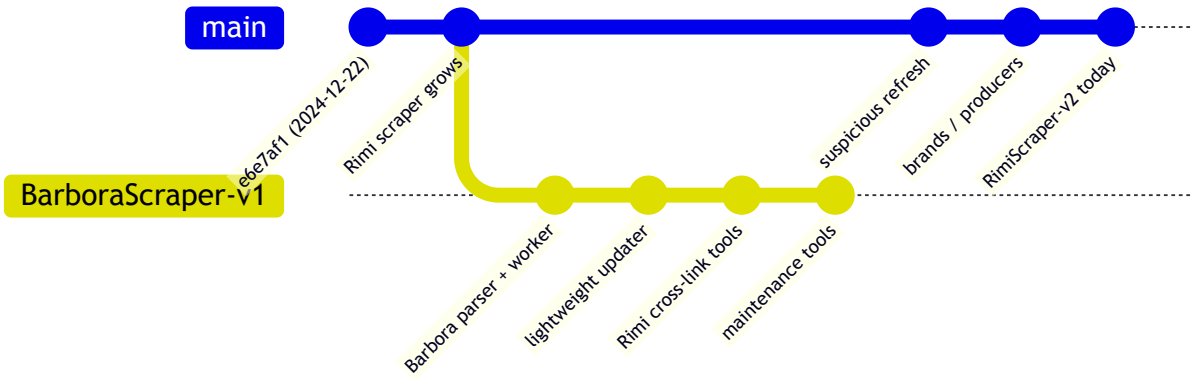
System overview



Data moves in one direction: retailer website → scraper → PostgreSQL → website. The website never writes to the scraper databases. The scrapers never call the website.

How the repositories relate

1. Shared code: the Barbora scraper is a fork



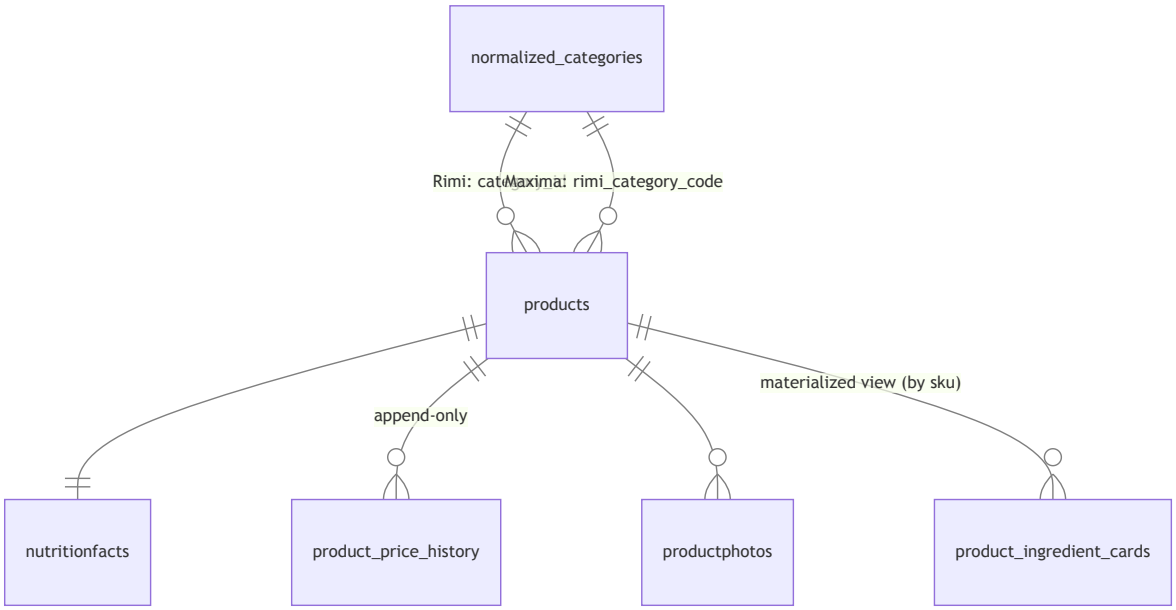
Both scraper repositories have the same root commit, `e6e7af1`. The fork kept most of the Rimi code: the job queue tables, `nutriscore_calculator.py`, `fiber_estimator.py`, the ingredient pipeline and the Streamlit layout. It then added Barbora-specific modules. Since the fork, each repository has changed on its own:

- **Rimi** added heartbeat leases for jobs, `suspicious_refresh` jobs and the nutrition validator.
- **Barbora** added the lightweight listing updater, stricter NOVA rules and the plan → preview → apply maintenance tools.

The two copies aren't kept in sync, so a fix in one doesn't automatically reach the other.

2. Shared data contract: same schema, two databases

Both scrapers write the same core tables. The website's Drizzle mappings in `ABCDE-Web-EU/lib/db/src/server/remote-sch` depend on them:

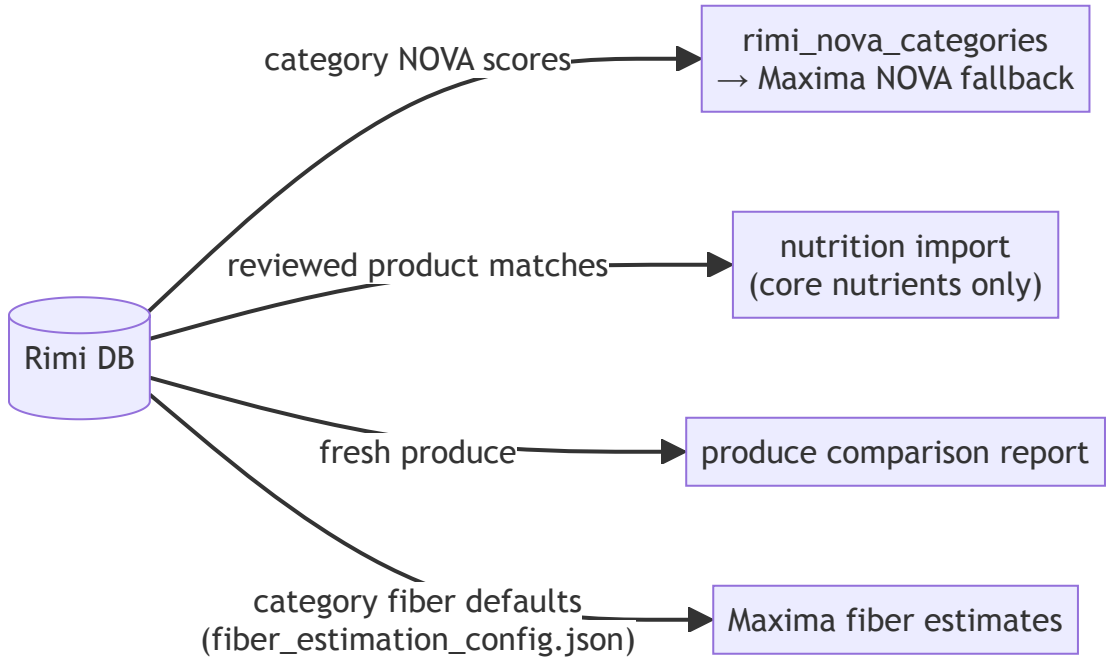


Contract	Rimi DB	Maxima DB	Website use
`normalized_categories`	**The category tree**	Barbora's own tree, not used by the site	All browsing and filtering
`products.category_id`	Rimi category	Barbora category	Rimi filtering
`products.rimi_category_code`	Text code, e.g. `SH-11-9-26`	**Integer = Rimi `category_id`**	Places Maxima products in Rimi's tree
`nutritionfacts.nutri_score_grade`	Calculated by the scraper	Calculated by the scraper	Grade badges, "healthier alternatives"
`product_price_history`	Append-only	Append-only	Latest row = current price (`LATERAL` join)
`product_ingredient_cards`	Materialized view	Materialized view	Ingredient breakdown on product pages
`products.is_hidden`	Set by the scraper UI	Set by the scraper UI	**Not checked.** The API never reads this column.

`rimi_category_code` means different things in the two databases: a text code in Rimi, a foreign key to a Rimi category in Maxima. Keep that in mind when changing either scraper's category logic.

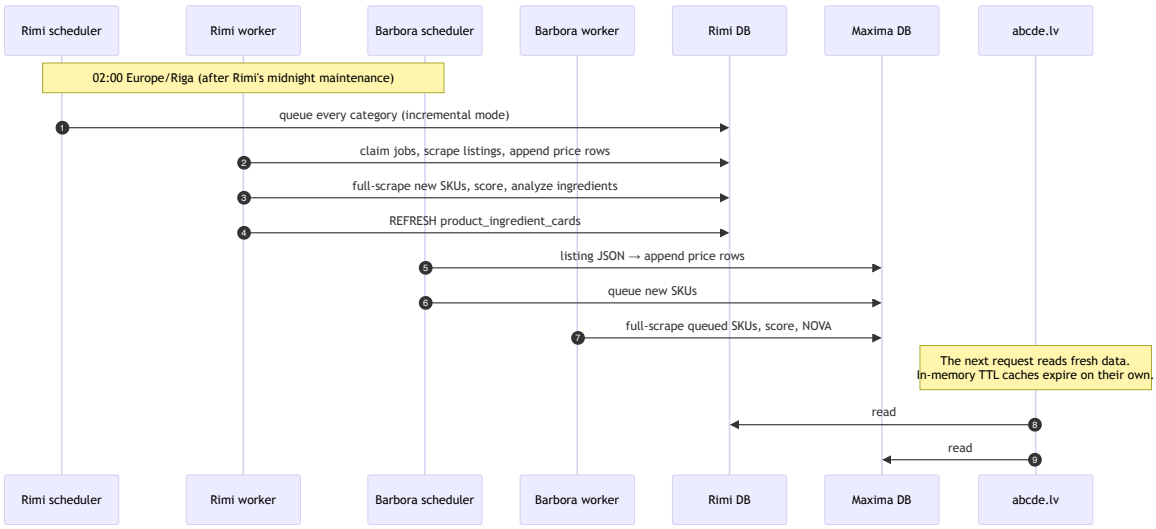
3. Data from Rimi used by the Barbora scraper

The Barbora scraper uses data from the Rimi database through `RIMI_DATABASE_URL`, opened read-only:



So the Maxima data partly depends on the Rimi data. Changing a NOVA score on a Rimi category changes the NOVA fallback for Maxima products the next time `sync_rimi_nova_categories.py` runs.

Daily cycle



Scoring is done in the scrapers

The website shows `nutri_score_grade` and `nova_score` exactly as the scrapers store them. It doesn't recalculate them. The algorithms are implemented in Python, once in each scraper:

- **Nutri-Score 2024.** Each nutrient earns step-function points $\text{pts}(x, T) = |\{i : x > t_i\}|$. The score is $S = N - P$, where N counts energy, sugars, saturated fat and salt, and P counts protein, fiber and fruit/vegetables/legumes. There are separate rules for beverages, cheese, red meat, and fats/oils/nuts. The derivation is in RimiScraper-v2/ARCHITECTURE.md.
- **NOVA.** Any ingredient flagged *avoid* or *limit* gives group 4. Otherwise the product gets its category's group. Barbora adds checks so that missing or contradictory analysis leaves the score unknown instead of guessing; see BarboraScraper-v1/ARCHITECTURE.md.

The two NOVA implementations have drifted apart. The same product can get a different NOVA group depending on which store sells it.

External services

Service	Used by	Purpose	Config
Google Gemini 2.5 Flash	both scrapers	Ingredient classification, translation	`AI_INTEGRATIONS_GEMINI_*`
USDA FoodData Central	both scrapers	Missing nutrition	`USDA_API_KEY`
Google Translate	both scrapers	Product name translation	`GOOGLE_TRANSLATE_API_KEY`
Google Cloud Vision	website	OCR for photo scans	`GOOGLE_CLOUD_API_KEY`
abcde-mcp (Replit)	website	MCP server reached through the `/mcp` proxy	fixed upstream URL
Neon	all	PostgreSQL hosting	`*_DATABASE_URL`
Replit	all	Hosting (Autoscale UI; Reserved VM planned for workers)	`.replit`

Risks across the repositories

1. **Schema changes need coordination.** Changing a shared table in either scraper can break the website's Drizzle mappings without any error until a page fails. There is no shared schema package or contract test.
2. **The forked code drifts.** Fixes to shared modules such as Nutri-Score, fiber and ingredient processing must be applied twice.
3. **The website depends on Rimi's category tree.** Maxima products appear in the site only if they have a `rimi_category_code`. Any change to Rimi's tree affects both stores.
4. **`is_hidden` has no effect on the site.** Both scrapers have an `is_hidden` flag meant to hide products from the browser UI. The website's API doesn't read it (searching `artifacts/api-server` and `lib/db` finds no references), so products marked hidden in the scrapers still appear on abcde.lv.
5. **The workers have no always-on host yet.** Both scrapers expect a Reserved VM. Until then the workers run on Autoscale with small chunks, so a large backlog can delay data on the site.